SIP MACHIGURUMI SIMULATOR · PROGRESS SEMINAR · 23 OCTOBER 2026

Activity-based travel demand modelling with a dynamic discrete choice model

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draft, 22 October 2026

In one paragraph

We are building a model of how each resident of Higashi-Hiroshima plans a whole day of activities and travel, learned from local travel diaries. It is designed to answer questions such as "if a BRT opens, how will behaviour and accessibility change, and for whom?". Its core idea is to **manage constraint heterogeneity properly**: every person keeps their own constraints (working hours, a child to collect, no car), so that what people *cannot* do is never mistaken for what they *do not want*, and this is done for a whole city at a cost that stays manageable.

1 The question

Higashi-Hiroshima is trying out new mobility, such as an autonomous, platooning BRT. Before a new line or mode is built, a planner wants to know how people will respond, and the response is more than a change of mode:

- **Mode:** who switches to the bus, and from which mode?
- **Timing:** do people leave home or work at different times?
- **Destination:** do they shop or spend free time in different places?
- **Activities:** do they add a stop, combine trips, or drop one?
- **Accessibility:** what can each person now reach and do, and who is left out?

These answers are connected, and they differ from person to person, so a model has to predict them together, for each person. A parent who collects a child at half past six, a commuter without a car and a retired resident will respond differently to the same BRT line, mostly because their **constraints** differ, not because their tastes do.

2 Why an activity-based model

A **travel demand model** describes how the people of a city travel: who makes a trip, to where, at what time, and by which mode. It is learned from real travel data, and its main use is to test a change to the city before anything is built.

A **trip-based model** predicts each trip on its own. An **activity-based model** starts from what people do during the day (work, shopping, collecting a child) and treats trips as the way of getting between those activities. Because the trips are linked, a change to one part of the day can move the rest of it: leave work at 18:30 and the evening shop may no longer fit; drive to work and the car is there for the shop on the way home.

3 What a dynamic discrete choice model is

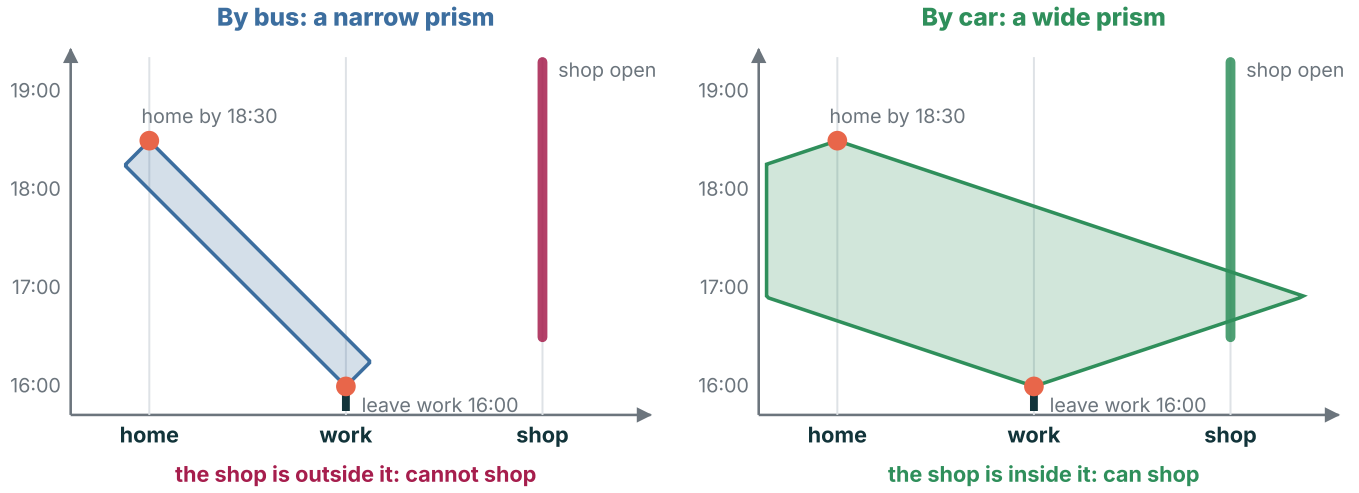
A **discrete choice model** describes a person choosing one option from a list, such as car, bus or walking. Each option gets an attractiveness score from its features, such as travel time and cost. More attractive options are more likely to be chosen, but never certain, because people differ in ways the data cannot see.

A **dynamic discrete choice model (DDCM)** applies this at every moment of the day, and values each choice by how attractive it is now *and* by what it leaves possible later: staying late at work may rule out the evening shop.

The model is used in two steps. **Estimation** learns people's preferences, such as how much they dislike travel time, from the diaries. **Simulation** then produces a schedule for every resident, for the city as it is or with a policy in place. Preferences stay fixed; only the city changes between scenarios.

4 The problem: people face different constraints

Constraints are the limits on what a person can do: fixed working hours, a child to collect, shop opening hours, whether a car is available. Hägerstrand's **space-time prism** draws them as every place a person can reach between two fixed appointments, given how fast they can travel. Two people in the same city, with the same shops and the same buses, can have very different prisms.



Time runs up, place runs across. The person is at work until 16:00 and must be home by 18:30 to collect a child. The shaded area is everywhere they could be in between. By bus the prism is narrow and the shop is outside it; by car it is wide and the shop is inside it. Same city, same shop, different possibilities. Illustrative.

Ignore constraints, and they read as preference. A diary records what people did, not why. Suppose a model does not know about these constraints and sees a parent who never shops in the evening, because of a pick-up at 18:30. The only explanation it has is that the parent dislikes evening shopping. The error spreads: if shops stay open later, the model predicts no change for that parent, who could in fact now shop after the pick-up, and its accessibility gain is missed. **This is why constraints matter for welfare:** accessibility (the logsum, in yen) is the value of all the days a person can still live, so misread constraints give the wrong winners and losers of a policy.

Why it is hard: the calculation is heavy

Because each choice is weighed against everything it leaves possible later, a DDCM has to value every situation a person could be in during the day: millions of them. Estimation repeats the whole calculation for every try of the preference numbers, hundreds of times.

Exact constraints for every person are heavier still

The fix, every person's own constraints, has a cost. The same shop at 18:00 is worth a lot to someone with a car, nothing to a parent with a pick-up, and less to a retired bus user. What an option is worth depends on what the person can still do afterwards, so each person needs a full calculation of their own. Models therefore usually simplify: fewer places, coarser zones and time steps, or the same constraints for everyone, which removes exactly the differences a planner needs.

5 Where the research stands

Managing constraint heterogeneity properly needs each person's own constraints, represented exactly, for a whole population. Recent research has achieved each half, but separately.

Approach	Each person's own constraints, exact?	Works for a whole population?
Västberg et al. 2020; Naqavi et al. 2023	yes	one calculation per person
Oyama & Hato 2019	one prism, for one origin and destination	yes, one shared map
Yao & Zhang 2026 (preprint)	differs only between a few groups	yes
This work	yes	yes, and estimated from diaries

6 Objectives

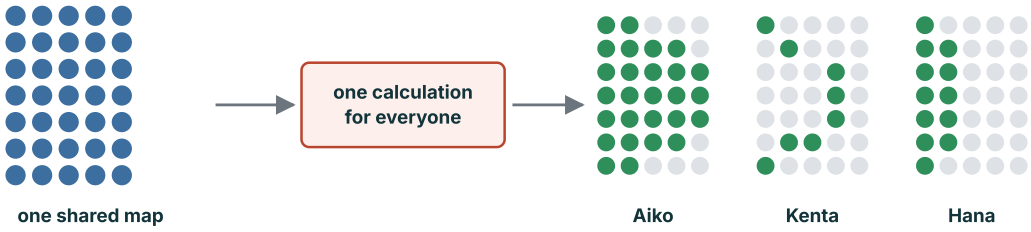
1. **Model each person's whole day**, estimated from local diaries, with every person's constraints exact. *(§10)*
2. **Manage constraint heterogeneity** at city scale: own constraints for everyone, at the cost of a few shared maps. *(§7–8)*
3. **Show what a policy does** to each person's activities, timing, destinations, modes and accessibility. *(§10: the BRT, city-wide policies, who gains)*

7 Our approach: managing constraint heterogeneity properly

The model writes the chance of each choice as two separate parts: whether the person **can** do it at all (their constraints, different for everyone) and how much people **like** it (preferences, shared and learned from the diaries). Because the two are kept apart, a "cannot" is never read as a "does not like".

A **constraint** is a rule. It looks at who the person is and the situation they are in, and answers one question about a possible next step: is it allowed, yes or no? A person's prism is everything their rules allow. We propose four types of constraint, sorted by what the rule depends on; the type decides how the model handles it.

Type	The rule depends on	For example	What the model does
Institutional the same for everyone	nothing about the person	opening hours, the network, the day ends at home	removed once, for everyone
Capability the means a person has	a coarse attribute	car ownership, which modes are available	one of a few shared maps
Anchor where and when one must be	a detailed attribute	home, workplace, work and school hours	a mask on the shared map
Interdependent set by another person	another person's choice	the household car, who collects the child	open: needs a household model



Each person's prism is the shared map with their own impossible options switched off (green = can, grey = cannot). Nothing is approximated: the result is the same as calculating every person separately, but the calculation runs once per shared map. However many people there are, the number of shared maps stays small: for example, one for people with a car and one for people without. Shapes are illustrative.

8 States, actions and data

Each moment of a person's day is a **state**: the time, the place, the activity, how long they have been at it, how they got there, and what is already done. An **action** is what they do next: stay, or travel somewhere for an activity by some mode. In Higashi-Hiroshima the model uses 15-minute steps, 144 zones, four activities and eight modes.

Preferences are estimated from the Higashi-Hiroshima weekday travel diaries, with door-to-door travel times, parking fees and real opening hours for every zone.

6,451
persons aged 13+ in the weekday diaries

70
preference parameters, estimated together

144
zones, door-to-door bus and rail times

9 What is new, and what is not

Built on others' work: the space-time prism (Hägerstrand 1970); a dynamic activity model with exact constraints and accessibility read from it (Västberg et al. 2020; Naqavi et al. 2023); a prism mask inside the calculation (Oyama & Hato 2019); the perturbed-utility way of writing the model (Yao & Zhang 2026).

New in this work:

- Every person's own constraints, kept exact, for a whole city, and **estimated** from real diaries.
- A **classification of constraint types** by what each rule depends on, which decides how each is handled.
- Naming the case that does not reduce, a constraint set by another person's choice, as the open limit.

Words used

Accessibility	The value of everything a person could do in the day, given their constraints; it can also be written in yen.
Preference	How much people value something, such as how strongly they dislike travel time. Shared, learned from diaries.
Shared map / mask	A map of options calculated once for many people; a mask switches off one person's impossible options on it.
MATSim	A traffic simulation that moves every trip through the road and bus network and returns travel times.

10 Results and policy use

Current specification (22 October), one iteration that estimated successfully, not the final model; choices such as ages 13+ and the reference parameters handle the data for now. 70 of 78 parameters estimated together on 6,451 persons, 63 significant at 5%, converged. Simulations: the full sample of 6,477 persons, three days each. Standard errors are robust (sandwich); scenario effects are point estimates.

What the model learned: value of travel time

Mode	Yen per minute	90% range (robust)
Car (driver)	79	62–106
Walk	65	52–85
Bicycle	48	38–64
Bus and train	23*	15–36

*Assumed: bus and train times are too noisy for the full model, so the time coefficient comes from a trip-level model on the same diaries (20.4 yen/min there, within the national rail range 15.2–37.5; 23 against time at home).

Does it reproduce observed days?

	Diary	Model
Mode shares (5 modes)	all within 1.2 points	
Trips per person-day	1.68	1.50
Average trip	36.6 min	31.2 min
Leisure trips by area	within about 1 point	
Shopping trips to other cities	4%	18%

The rhythm of the day and the work peak match. The model makes fewer, longer outings, and its destinations count shops, not their size.

What the policies do

- **BRT**: small on average (most trips are by car) but concentrated: workers +2.16 and people without a car +2.00 yen a day; it works as a feeder to Saijo station, shortening train trips too (349 zone pairs, median 3 min), so most new public transport use is by train. Faster buses only (headway 6 / 15 / 12 min, bus fare): no road space, congestion, comfort or reliability yet.
- **Cheaper or faster PT**: more trips, mostly new trips to Hiroshima by train; the corridor's share falls, but its own visits barely change.
- **Parking ×2**: the three central zones lose about 34 visits a day, and most of those outings are not made at all.

BRT: who gains (yen per person-day)

Group	Time saved	Logsum
Workers	+2.16	+1.56
No car in the household	+2.00	+1.57
Corridor residents	+1.46	+1.16
Elsewhere	+0.86	+0.66
Car owners	+0.56	+0.44
Non-workers	+0.29	+0.32
All persons	+1.06	+0.83

Corridor: home within 1.5 km of a BRT stop. 3.1% of persons gain, some by over 100 yen a day.

Policy scenarios (same people, preferences held fixed)

Scenario	Access ¥/day	Gain / lose	Car Δpt	Bus+train Δpt	Trips/day
BRT corridor	+1.06	3% / 0%	−0.03	+0.04	+0.0%
PT 20% faster	+61	38% / 0%	−2.22	+2.31	+1.3%
PT half fare	+93	100% / 0%	−3.22	+3.49	+2.3%
Parking +50%	−3.9	0% / 66%	−0.17	+0.08	−0.4%
Parking ×2	−6.6	0% / 66%	−0.29	+0.14	−0.7%
Work 1 h later*	−532	4% / 37%	+0.98	−0.74	+0.8%

PT = bus and rail. Access: time saved × 20.4 yen per minute for BRT and faster PT, the logsum otherwise. *Sensitivity test: read the direction, not the size. Congestion scenario not run.

How days and destinations change (against today)

Scenario	Trips per day	Shop + leisure	Average trip	Minutes travelling	Free-time: corridor Δpt	other cities Δpt
BRT corridor	+0.0%	+0.1%	−0.1%	+0.3%	+0.01	+0.01
PT 20% faster	+1.3%	+4.0%	−4.0%	−2.8%	−0.98	+1.94
PT half fare	+2.3%	+6.8%	+4.2%	+7.0%	−2.15	+3.91
Parking +50%	−0.4%	−1.4%	−0.1%	−0.4%	−0.55	+0.23
Parking ×2	−0.7%	−2.4%	−0.0%	−0.5%	−0.94	+0.39
Work 1 h later*	+0.8%	+4.6%	−1.4%	−0.5%	−0.13	+0.47

Per person-day, full sample. Today: 1.53 trips, 0.36 of them shopping or leisure, 31.2 minutes per trip, 46.7 minutes travelling; 46% of free-time trips go to the corridor, 16% to other cities.

11 Where it fits in the Machigurumi simulator



The parts are still developed separately. If integrated, the DDCM learns from the survey app's schedules, produces a schedule for every resident from population synthesis, exchanges trips and travel times with MATSim, and tells co-location who is where and when. Later, co-location can feed back co-activities (who is with whom) as coupling constraints (dashed).

First test: the model's day plans in MATSim

The day plans of 19,350 simulated person-days (a 9% sample) were loaded onto MATSim's road network: routes re-optimised; mode, time and destination stay the model's; one pass, no feedback loop. The traffic is plausible: morning and evening peaks, and the main roads carry the most cars (link volumes correlate 0.79 with an earlier full-population run). Car travel times from this run match the diary with almost no bias (−0.06 min, against −6.2 for the static travel times). With the BRT, the BRT-scenario plans give about 3,330 BRT boardings a day, 4% more than today's plans on the same network (busiest stop: Saijo station), and car veh-km fall by 0.18%. This is a lower bound: MATSim's public transport carries only about a third of the model's bus and train trips (1,536 of 4,421; no stop within 800 m for the rest). Congestion levels are not usable from a 9% sample; next are the full 214,000-person population, a public transport setup matching the model's door-to-door times, and the feedback loop for BRT lanes and congestion.

12 What it cannot do yet

- One weekday per person: no week, habit or learning over time.
- Work duration and start time are taken from the diary, not chosen.
- Individuals, not households: shared cars and joint activities are not yet coupled.
- Road times are static unless the MATSim loop is run.
- The same preferences for everyone in the current estimate; differences come from constraints and observed attributes.
- Destinations count shops, not their size, and the store data (opening hours, shop counts) is incomplete, so shopping is spread over too many zones (about 110 a day against 70 in the diary); outings come out too few and too long.
- The BRT is faster buses only: no lane, congestion response, comfort or reliability yet. The BRT is part of the bus network: a BRT-only trip is a bus trip, and as a feeder to rail the trip counts as train, its main mode. Mode shares therefore cannot show BRT use (only MATSim counts boardings); the mode specification will be revisited. Public transport's time value is assumed.

Next. Because the BRT changes only travel time, its effect is small; the city-wide levers show the model does respond. Next: a richer specification (modes, store size, what the BRT offers beyond time) and a closed loop with MATSim (lanes, congestion, the full population).

13 Key messages

1. Trips are **linked** through activities, and policies act on the whole schedule.
2. Constraint heterogeneity is **managed**, not ignored: every person keeps their own constraints, so changes in behaviour and accessibility can be read person by person.
3. The model is **estimated** from local diaries and is designed to feed MATSim and the rest of the simulator.

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